

Scientific computing and data visualization in Python

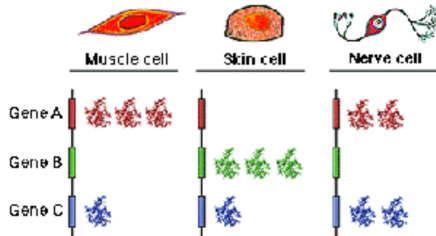
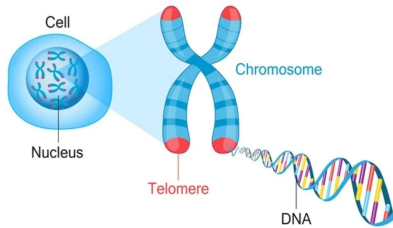
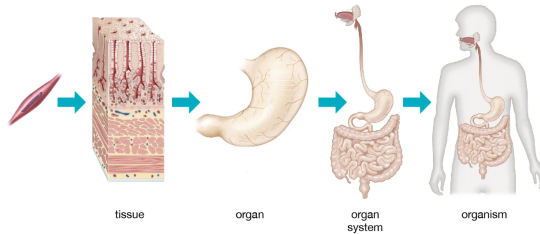
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2026

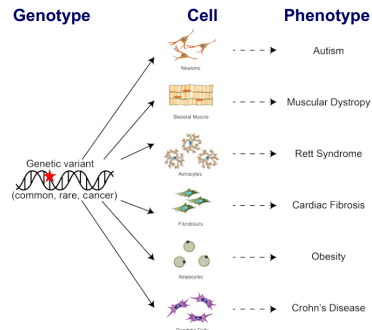
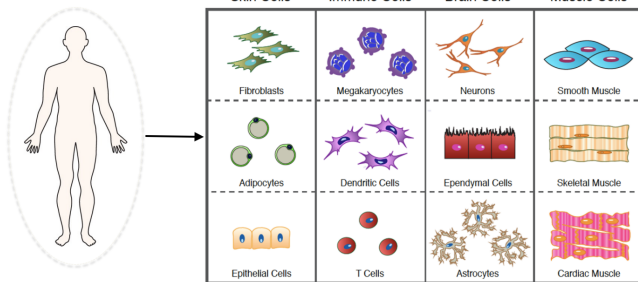
Levels of organisation

Cells are basic units in organisms.



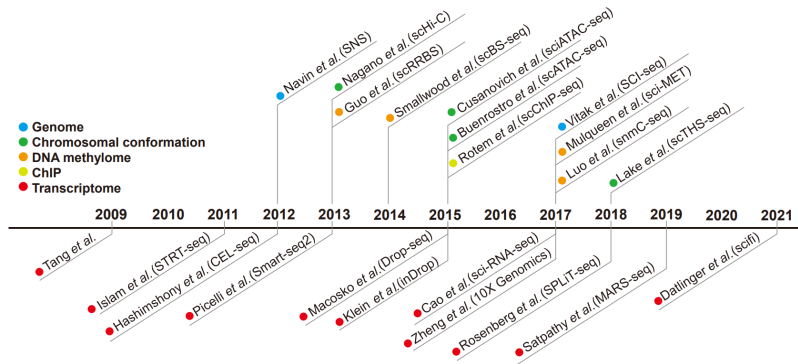
cell types

Cells are our core constituents. They are usually classified by characteristic molecules, structures, and functions.



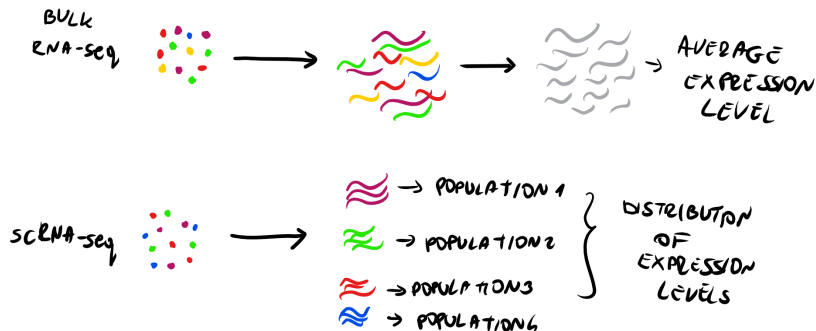
Cells are key intermediate from genotype to phenotype, e.g., essential functional dissection of genetic variants.

Timeline of technical progress of single-cell analysis.



Single-Cell Analysis Using Machine Learning Techniques and Its Application to Medical Research, biomedicines, 2021

Bulk vs scRNA-seq



Bulk RNA-seq - applications:

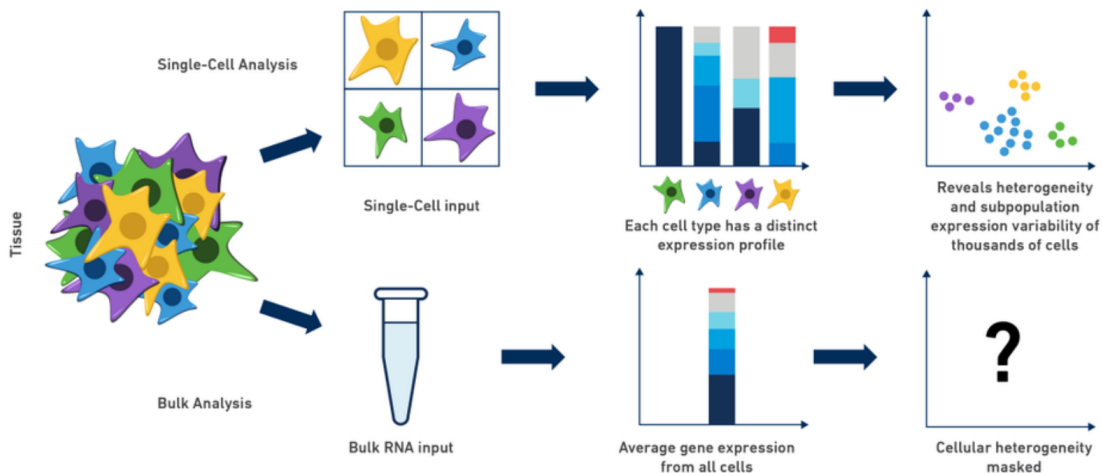
- comparative transcriptomics
- disease biomarkers

Suitable for homogenous systems (insufficient for studying heterogeneous systems).

scRNA-seq - applications

- define heterogeneity
- identify rare cell populations
- cell population dynamics

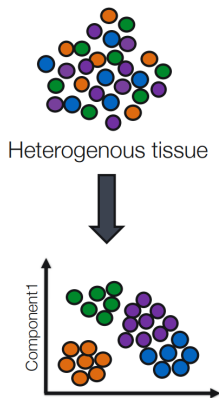
scRNA-seq - heterogeneity



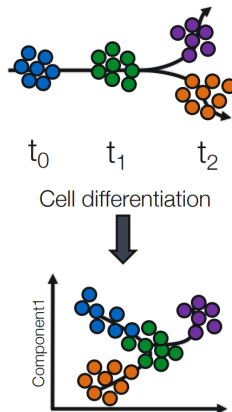
<https://www.10xgenomics.com/community-redirect>

scRNA-seq - two example applications

Studying heterogeneity



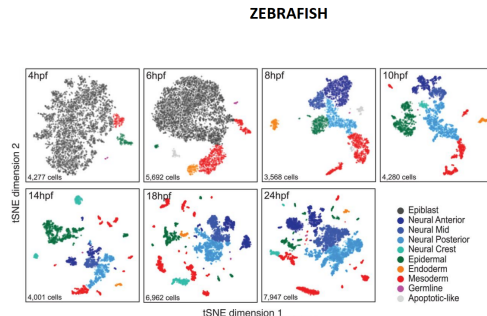
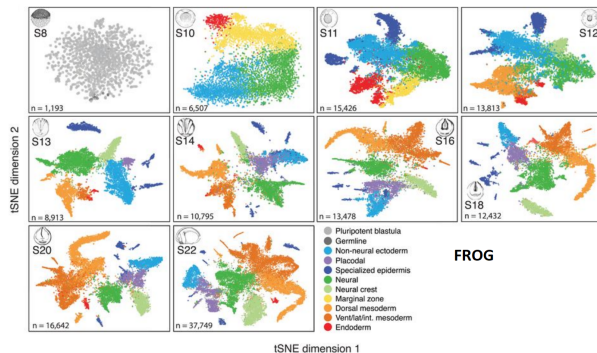
Lineage tracing study



Liu S and Trapnell C. Single-cell transcriptome sequencing: recent advances and remaining challenges, F1000 Research 2016 (doi: 10.12688/f1000research.7223.1)

Junker and van Oudenaarden; Every Cell Is Special: Genome-wide Studies Add a New Dimension to Single-Cell Biology, Cell 2014 (doi: 10.1016/j.cell.2014.02.010)

scRNA-seq - lineage

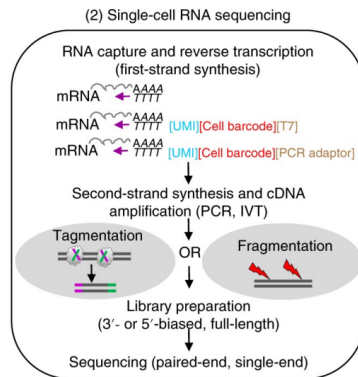
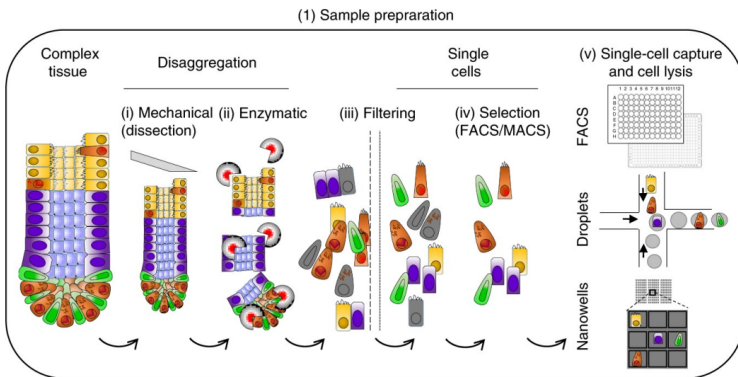


JA. Briggs et al. The dynamics of gene expression in vertebrate embryogenesis at single-cell resolution, Science 01 Jun 2018 (DOI: 10.1126/science.aar5780)

DE Wagner et al. Single-cell mapping of gene expression landscapes and lineage in the zebrafish embryo, Science 01 Jun 2018 (DOI: 10.1126/science.aar4362)

scRNA-seq - steps

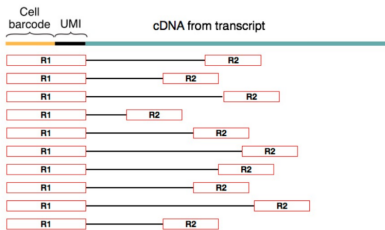
- **Step1: Sample preparation** - cells are physically separated into a single-cell solution from which specific cell types can be enriched or excluded
- **Step2: Single-cell RNA sequencing**
- **Step3: Data processing**
- **Step4: Data analysis**



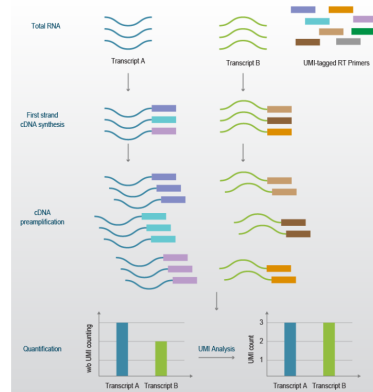
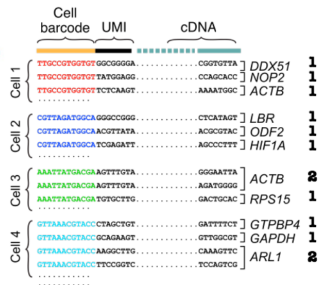
Step3: Data processing

UMIs (Unique molecular identifiers) - short (4-10bp) random barcodes added to transcripts during reverse-transcription.

Biased paired-end reads



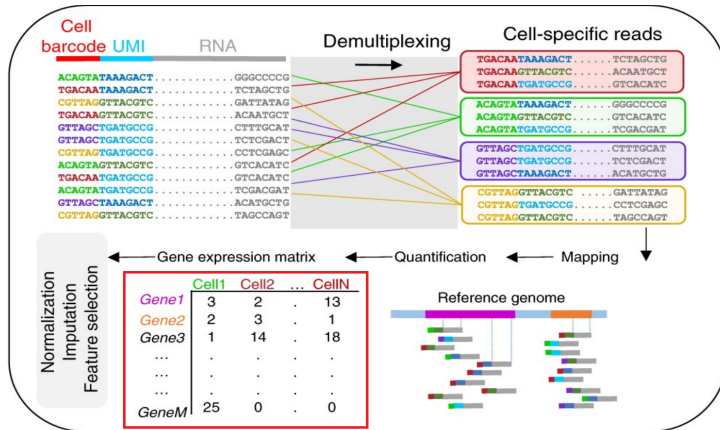
Grouping barcodes to assign reads to cells



- They enable sequencing reads to be assigned to individual transcript molecules and thus the removal of amplification noise and biases from scRNA-seq data.
- They reduce the amplification noise by allowing (almost) complete de-duplication of fragments.

Step3: Data processing

In each gene, within each cell, the total number of unique UMI is counted and reported as the number of transcripts of that gene for a given cell. Finally, we obtain gene counts for every cell. Usually, before the analysis we normalize these counts and transform values using logarithm.



Lafzi et al. Tutorial: guidelines for the experimental design of single-cell RNA sequencing studies, Nature Protocols 2018 (<https://doi.org/10.1038/s41596-018-0073-y>)

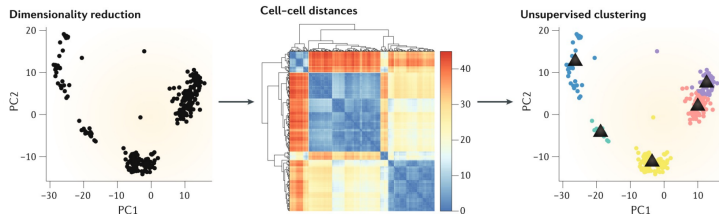
Step4: Data analysis

We also should give attention to drop-outs.

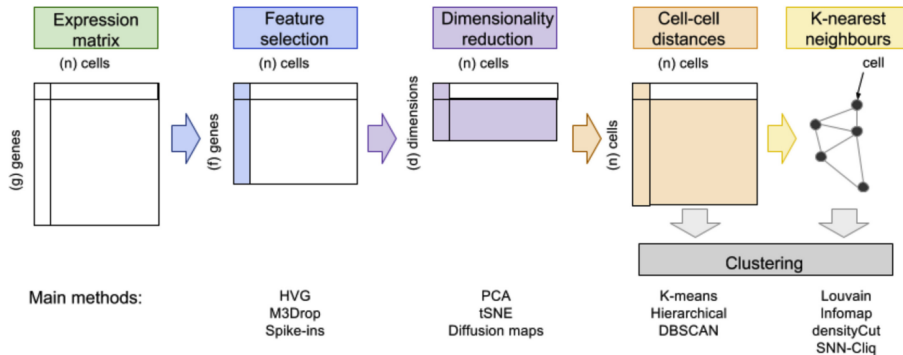
- **Drop-outs** - a gene is observed at a moderate or high expression level in one cell but is not detected in another cell. Why? technical artifacts, cell type differences, biological factors...

Data analysis

- Dimensionality reduction: Linear (PCA), Non-linear (t-SNE, UMAP).
- Cell clustering (k-means, hierarchical clustering, graph-based clustering).
- Trajectory analysis (To understand cellular differentiation and evolution over time. It involves reconstructing and visualizing developmental trajectories of cells based on gene expression data. This technique allows for identification of developmental lineages, determination of when cells initiate differentiation, and identification of genes regulating this process.)



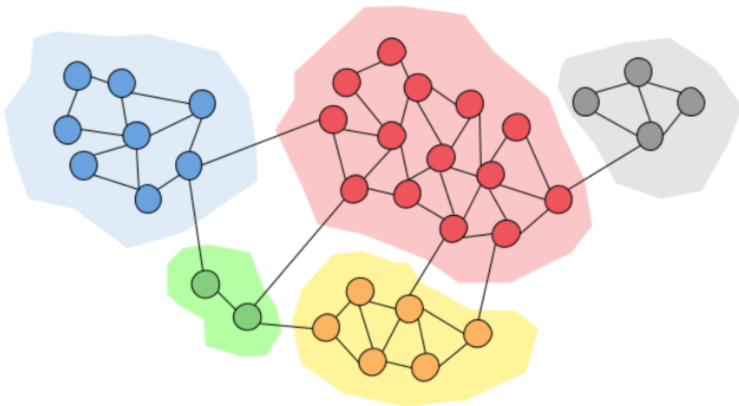
Step4: Data analysis



- Feature selection and dimensionality reduction extract the most informative genes and represented features from background noise, respectively.
- Cell-cell distances are then calculated in the low dimensional space for clustering cells to clusters.

Clustering methods

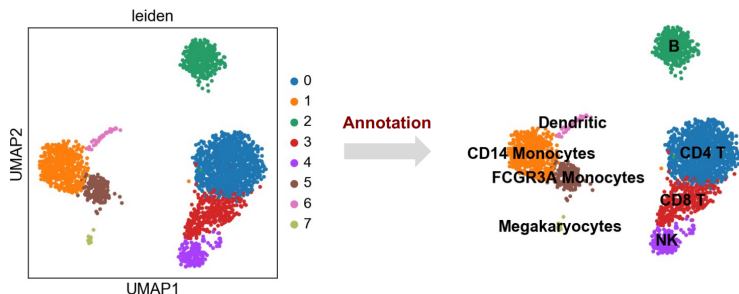
- K-means,
- Hierarchical clustering,
- Graph-based clustering (there, node = cell, edge = similarity between two cells)



Cell type annotations

The aim is to assign the cell type for each cell (cell types → cellular functions)

- **Supervised methods:** a training dataset labeled with the corresponding cell population is needed to train the classifier.
- **Prior-knowledge based methods:** either a marker gene file is required as an input or a pretrained classifier for specific cell populations is provided.



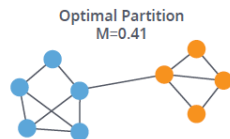
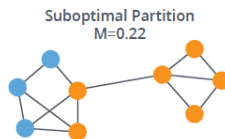
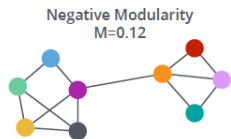
Graph-based clustering

Modularity - measures the density of links inside communities compared to links between communities. For a weighted graph

$$Q = \frac{1}{2m} \sum_{ij} [A_{ij} - \frac{k_i k_j}{2m}] \delta(c_i, c_j)$$

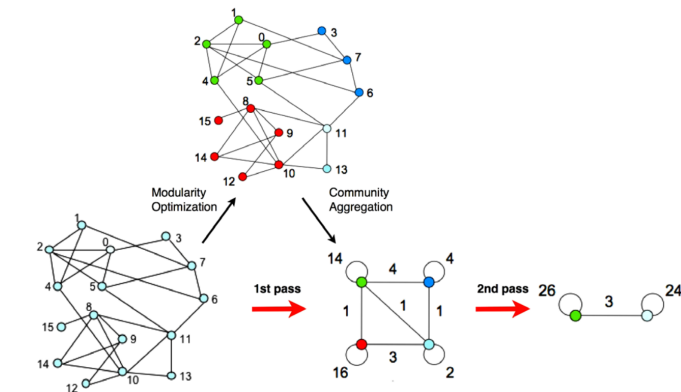
where

- A_{ij} represents the edge weight between nodes i and j ,
- k_i and k_j are the sum of the weights of the edges attached to nodes i and j , respectively,
- m is the sum of all of the edge weights in the graph,
- c_i and c_j are the communities of the nodes,
- δ is Kronecker delta function ($\delta(x, x) = 1$, $\delta(x, y) = 0$).



Clustering goal: assign each node a module (a group of highly connected nodes) to maximize modularity as an objective function.

Louvain clustering



Paper: Fast unfolding of communities in large networks

Two phases

- modularity is optimized by allowing only local changes of communities
- the communities found are aggregated in order to build a new network of communities

These two steps are repeated iteratively until no increase of modularity is possible. Advantages: intuitive, easy to implement, unsupervised, extremely fast.

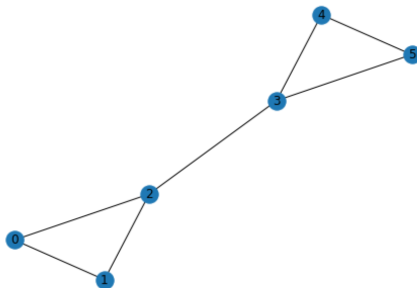
Louvain clustering - algorithm

- We start with a weighted network of N nodes.
- Firstly, we assign a different community to each node of the network - there are as many communities as there are nodes.
- Then, for each node i we consider the neighbours j of i and we evaluate the gain of modularity that would take place by removing i from its community and by placing it in the community of j .
- The node i is then placed in the community for which this gain is maximum (if no positive gain is possible, i stays in its original community). Remark: Node may be, and often is, considered several times.
- The first phase stops when a local maxima of the modularity is attained (no individual move can improve the modularity).
- The second phase of the algorithm consists in building a new network whose nodes are now the communities found during the first phase.
- The weights of the links between the new nodes are given by the sum of the weight of the links between nodes in the corresponding two communities.
- Links between nodes of the same community lead to self-loops for this community in the new network

The Leiden method is an improved version of the Louvain method that provides better performance in larger networks. Both methods cluster nodes in network based on maximizing modularity, but differ in their approach.

- **The Louvain community detection method** operates iteratively to optimize modularity by assigning nodes to clusters hierarchically. In each iteration of the algorithm, nodes are assigned to clusters based on maximizing modularity, and then new clusters are formed by grouping clusters from the previous iteration. The algorithm stops when modularity no longer increases or a maximum level of hierarchy is reached.
- **The Leiden community detection method** introduces improvements to the Louvain algorithm that allow for more efficient clustering in larger networks. Instead of iteratively assigning nodes to clusters, the Leiden method **uses a recursive algorithm** that replaces smaller clusters with larger ones if it leads to improved modularity. This allows for more precise results in less time.

- Consider the following graph



Calculate its modularity assuming one and two communities.

- Open *scRNAdata.csv* file. Rows are genes, columns are cells.
 - How many cells and genes were considered?
 - Standardize the data and apply PCA with $k = 2$.
 - Plot the PCA1 vs PCA2 components. Do you notice some groups?
 - Build a graph on cells, where the edge weight is defined as $\exp(-||c_i - c_j||)$.
 - Find the best partition of the graph.
 - Plot the PCA1 vs PCA2 components once again. Colour the data points according to partition.

Go to https://singlecell.broadinstitute.org/single_cell, search for **Single-cell transcriptomics of the spinal cord of a severe SMA mouse** and try to answer the following questions:

- What was the aim of the experiment? (Summary)
- How many cells were investigated?
- How many categories were considered? Which of them was the most abundant?
- Consider the SMN1 gene. What is the function of the gene? On which chromosome it is located? Is it expressed among all considered cell types? Where the expression is the highest?
- Consider the BICD2 gene. What is the function of the gene? On which chromosome it is located? Is it expressed among all considered cell types? Where the expression is the highest?
- Give an example of gene which is prevalent in blood cells, but not in the remaining ones.

Consider the paper **Single-cell transcriptome identifies molecular subtype of autism spectrum disorder impacted by de novo loss-of-function variants regulating glial cells**. Read at least the abstract and then, try to address the following questions:

- What type of cells were considered? How many clusters did the authors obtain? How did they identify differentially expressed genes? (Results/subsection: Construction. . . Page2)
- What are „Brain critical exons” and „pLI”? What was the purpose of applying these approaches? (Results/subsection: Identification. . . Page3)
- After enrichment analysis, which clusters were identified? (Results/subsection: Identification. . . Page3/Page4, Fig.1.)
- How did authors assign cluster identity? (Results/subsection: Subtypes of glial cells. . . Page5)
- Which cell type was overrepresented by its marker genes in cluster 13 in ACC? (Fig. 3A, Page5)